Nested evaluation of DNABERT and sequence composition for oncogene and tumor suppressor classification

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Working manuscript • 27 October 2026 • Not peer reviewed

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Abstract

Background: Pretrained DNA representations offer a way to model nucleotide context, but their value for distinguishing oncogenes from tumor suppressor genes must be assessed against simple sequence features and under evaluation that controls detected homology. We evaluated whether frozen or fine-tuned DNABERT representations improve classification from complete spliced coding sequences.

Methods: Historical COSMIC and OncoKB exports were audited to retain exclusive cancer-role labels. The resulting coding-sequence cohort contained 809 genes, comprising 441 oncogenes and 368 tumor suppressor genes. We used fixed five-fold outer and three-fold inner cross-validation under stratified gene splits and protein-similarity-grouped splits. Comparators were a class-prior predictor and logistic regression on length/GC features, 3-mer frequencies, 6-mer frequencies, or frozen DNABERT embeddings. Fine-tuning used a single learning rate and a two-epoch inner budget, requiring 40 fresh neural fits. Predictions were pooled at gene level. Uncertainty was estimated using 2,000 paired bootstrap resamples, conditional on fitted models and selected thresholds.

Results: Fine-tuned DNABERT achieved mean outer-fold balanced accuracy of 55.68% (95% interval 52.34–59.05%) on stratified splits and 53.31% (49.98–56.74%) on grouped splits. Frozen DNABERT achieved 56.44% and 56.16%, respectively. The best observed sequence-composition baselines achieved 59.95% for 6-mers on stratified splits and 56.50% for 3-mers on grouped splits. Fine-tuning minus 3-mer performance on grouped splits was −3.19 percentage points (95% interval −7.02 to +0.53). Verification of the saved evidence found no gene overlap between training and held-out partitions or detected-group overlap in grouped partitions.

Conclusions: This short fine-tuning protocol did not establish an advantage over simple sequence features. The findings support reporting strong composition baselines and explicit similarity-aware evaluation, while leaving broader tuning, repeated-seed stability, and alternative sequence inputs unresolved.

Keywords: DNABERT; oncogene; tumor suppressor; coding sequence; nested cross-validation; homology; sequence composition

Introduction

Cancer gene resources distinguish genes associated with oncogenic activity, tumor suppression, or more than one cancer-related role. COSMIC Cancer Gene Census provides curated cancer-gene annotations, including role categories [1]. OncoKB supplies curated biological and clinical information about cancer alterations [2]. These resources make a supervised sequence benchmark possible, but their labels are not a context-free description of every gene in every tumor. A binary experiment therefore needs an explicit eligibility policy and a clear statement of the population to which its results apply.

DNABERT learns contextual representations from DNA k-mer tokens through pretraining [3]. Such representations could potentially support downstream gene-role classification. That possibility does not establish that a pretrained encoder adds value over short-pattern composition for a small labeled cohort. A comparison with length, GC content, and k-mer frequencies can determine whether a more computationally demanding representation offers a measurable advantage within a specified training budget.

Evaluation also depends on what is held out. Stratifying genes by class preserves label balance, but related coding sequences may remain on both sides of a split. Similarity-aware partitioning changes the generalization question by withholding detected related groups. SpanSeq demonstrates the importance of considering sequence similarity during model development and assessment [4]. Here, we implemented grouping from protein alignments and connected components rather than using the SpanSeq executable.

We studied a closed-cohort task: distinguish exclusively annotated oncogenes from exclusively annotated tumor suppressor genes using one consistently selected complete coding sequence per gene. We compared four simple baselines with frozen and fine-tuned DNABERT using identical nested folds within each split setting. The objective was to measure the performance of these specified workflows and retain auditable evidence of their fitting boundaries. The experiment was not designed to predict cancer risk, discover cancer genes among all human genes, or assign clinical significance to individual variants.

Methods

Label sources and cohort construction

The label audit used recovered historical exports containing 758 COSMIC records and 1,192 OncoKB records. Exact file bytes were recovered from repository history; the original database releases and download dates could not be verified. The exports are therefore treated as a historical snapshot rather than a current database census. The audit preserved source annotations, stable identifiers, matching decisions, and exclusion reasons. Opposing exclusive annotations and unresolved identities were quarantined. Explicit dual-role genes were excluded from the binary task instead of forcing a single label.

The 1,950 source records resolved to 1,292 identity groups. Of these, 447 had an eligible exclusive oncogene label and 369 an eligible exclusive tumor suppressor label. The remaining groups comprised 125 explicit dual-role groups, 64 unresolved groups, and 287 groups without a target role. The 816 eligible genes form an exclusive-role union: matching support from both sources was available for 303, COSMIC-only label support for 167, and OncoKB-only label support for 346. Source membership alone was not counted as agreement.

Reference sequences and transcript policy

Sequences were obtained from GRCh38.p14 using the Ensembl release 114 archive. Stable source-provided Entrez identifiers were mapped to a unique primary nuclear protein-coding locus; symbols were not used to guess locus identity. Transcript selection prioritized MANE Select, followed by Ensembl Canonical, then longest translation with a stable-identifier tie-break. MANE supplies a jointly defined transcript resource from NCBI and EMBL-EBI [5]. Among retained genes, 808 selected transcripts were MANE Select and one, MUC16, used Ensembl Canonical.

The selected transcript’s complete spliced coding sequence was retained in transcript orientation, excluding introns and untranslated regions. Quality checks required nonempty ACGT sequence, length divisible by three, and agreement with annotated protein length allowing a terminal stop codon. The permitted ambiguous-base fraction was zero. Non-ATG starts were retained with flags rather than automatically excluded. A failed selected transcript was not replaced after its quality result was observed.

Seven eligible genes lacked usable CDS under this policy: four lacked an archived mapping, two had ambiguous primary coding mappings, and one failed the ambiguity rule. The analysis therefore retained 809 genes, including 441 oncogenes and 368 tumor suppressors. CDS lengths ranged from 135 to 43,524 bases, with a median of 1,764 and a mean of 2,516.5 bases. The cohort contained 2,035,842 retained coding bases.

Windows and gene representations

Each CDS was divided into balanced, nonoverlapping windows of at most 510 raw bases. For a sequence of length L, n = ceiling(L/510) windows used boundaries floor(iL/n) and floor((i+1)L/n). This covered every retained base exactly once and avoided prefix-only truncation. The 809 genes produced 4,389 windows; windows were computational units and did not count as additional labeled samples.

DNABERT inputs were overlapping 6-mer tokens prepared with the checkpoint's matching tokenizer and vocabulary. Encoded lengths were checked without truncation. Window vectors were the mean of final-layer hidden states over real tokens, excluding special and padding tokens. For a gene g with windows w, its representation was $z_g = \sum_w a_{gw} h_{gw}$, where h_{gw} is the token-mean window vector and a_{gw} is the window's base length divided by the full CDS length. The weights sum to one. Pooling covers the complete CDS but does not create attention between separate windows or preserve their global order.

Homology graph and fixed partitions

Protein translations from the selected transcripts defined groups; model inputs remained DNA. MMseqs2 version 18-8cc5c performed all-versus-all protein alignment [6], enumerating 809^2 ordered sequence pairs without a prefilter. An undirected edge was retained when either search direction met all three criteria: exact aligned identity of at least 30%, aligned-span coverage of at least 80% for each sequence, and E-value at most 10^{-5}. Identity used integer match counts divided by aligned length, including internal gaps. Connected components kept transitively related genes together.

The graph contained 205 edges and 673 components, including 586 singletons, with a largest component of 13 genes. Sixteen components contained both labels. The threshold describes broad near-full-length detected protein similarity, not all evolutionary or domain relationships. Sequence-only threshold diagnostics were recorded before model evaluation. In the stratified setting, 165 of the 205 qualifying edges crossed outer-fold assignments; none crossed in the grouped setting. This documents a structural difference between designs, not a measured causal amount of score inflation.

Five outer folds and three inner folds were fixed using seed 42. StratifiedKFold defined the gene setting and StratifiedGroupKFold defined the grouped setting. Both labels occurred in every training and held-out partition. All models used the same gene and inner/outer fold assignments within a setting. Derived windows followed their gene. Outer-test data did not enter preprocessing, class weighting, epoch selection, regularization selection, or decision-threshold selection.

Classical baselines and frozen representations

The class-prior predictor estimated class proportions from the current training partition. Length/GC logistic regression used the natural logarithm of complete CDS length and its GC fraction. The 3-mer and 6-mer baselines used complete fixed vocabularies of 64 and 4,096 features, respectively. Each feature was its overlapping full-CDS k-mer count divided by $L-k+1$. Reverse-complement features were not collapsed, and there was no feature selection or augmentation.

Logistic models used L2 regularization with liblinear, a fitted intercept, at most 3,000 iterations, and tolerance 10^{-6} in scikit-learn [7]. Standardization was fitted only on the current training genes. Class weight for class c was $N/(2N_c)$, calculated from those genes. Inverse regularization C was selected from 0.01, 0.1, and 1.0 by

mean inner-fold balanced accuracy at threshold 0.5; ties preferred lower C. A final threshold maximized balanced accuracy on complete selected-candidate inner out-of-fold predictions, with ties resolved by proximity to 0.5 and then the lower threshold. Final refits used all outer-training genes.

Frozen DNABERT used the pristine zhihan1996/DNA_bert_6 checkpoint at immutable revision a79a8fd96ad172f964a4dbef3f4d7545a5034baa. Model configuration, tokenizer, and weight provenance were recorded. The encoder ran without dropout in float32, producing 768-dimensional gene vectors. Weighted accumulation used float64 before conversion to float32. The same nested logistic procedure was fitted to these vectors; the encoder was not adapted to labels. Classical predictions were reused from the audited reference run, while frozen representations and their probe were produced in the recorded local run.

Neural fine tuning

Each fit began from the same pristine encoder with a fresh two-class linear head and fresh optimizer state. A legacy supervised checkpoint with unverifiable exposure history was excluded. This guards against unknown prior supervision for the present task; it does not imply that human reference sequences were absent from upstream unsupervised pretraining.

The fixed protocol used AdamW with learning rate 2×10^{-5}, weight decay 0.01, gradient clipping at norm 1, gene minibatches of four, and window microbatches of two. One class-weighted cross-entropy loss was calculated per complete pooled gene, normalized by the actual gene-minibatch size. The unused encoder pooler was frozen. A two-pass implementation first computed pooled vectors and then replayed window computations with the same dropout random-number state to propagate gradients while limiting memory.

Inner fits ran for two epochs. For each fit, the selected epoch maximized its validation balanced accuracy at threshold 0.5, with ties preferring the earlier epoch. Selected-epoch inner out-of-fold probabilities determined the outer threshold using the rule above. The outer model was trained from scratch for the ceiling of the median of the three selected inner epochs. No outer-test early stopping was used. Across both settings, the design required 30 inner fits and 10 refits, totaling 40 neural fits. The three-candidate logistic regularization search and the one-candidate neural learning-rate protocol are different tuning budgets; the comparison evaluates these particular workflows rather than equally optimized model families.

Metrics uncertainty and verification

Balanced accuracy, the mean of oncogene sensitivity and tumor suppressor sensitivity, was the primary metric. The primary aggregate was the unweighted mean across five outer-fold balanced accuracies, distinct from balanced accuracy computed after pooling all predictions. Secondary pooled metrics included AUROC, average precision for each label, Matthews correlation coefficient, macro-F1, per-class precision and recall, confusion counts, and Brier score. Oncogene was the positive class.

Percentile 95% intervals used 2,000 seeded bootstrap draws. Genes were sampled in the stratified setting and entire homology components in the grouped setting. The same sampled units were used for paired model differences, recomputing mean outer-fold balanced accuracy. The intervals condition on trained models, selected thresholds, and fixed folds; models were not retrained. They do not estimate seed-to-seed or full training-procedure variability. Comparisons were exploratory, without multiplicity correction, and five outer folds were not treated as five independent experimental replications.

The local numerical backend was Apple arm64 MPS with Python 3.12.16, PyTorch 2.9.1, Transformers 4.57.1, NumPy 2.3.5, SciPy 1.17.0, and scikit-learn 1.8.0. Deterministic algorithms were enabled. On-device checks recorded dropout-gradient replay, RNG restoration, padding invariance, encoder updates, and checkpoint round-trip behavior. Saved evidence was independently checked for artifact-hash agreement,

gene/fold coverage, partition disjointness, recorded-group isolation, training class counts and weights, fit lineage, threshold-consistent predictions, and recomputed mean fold balanced accuracy. CPU/MPS bitwise equivalence was not established.

Results

Completion and data integrity

All 40 neural fits completed. There were 30 distinct inner fits and 10 outer refits, and 1,618 fine-tuned outer predictions, one per gene in each setting. Across six models and two settings, the prediction table contained 9,708 records. All 15 artifact hashes in the final run manifest matched the submitted files. Saved gene assignments agreed across models, and grouped partitions had no overlap in their recorded homology components. The runner audit reported a passed status. Of 30 inner fits, 21 selected epoch 2 and nine selected epoch 1. Eight outer refits used two epochs and two used one epoch.

Table 1. Mean outer-fold balanced accuracy with conditional 95% bootstrap intervals.

Model	Stratified BA % [95% interval]	Grouped BA % [95% interval]
Class prior	50.00 [50.00, 50.00]	50.00 [50.00, 50.00]
Length and GC	55.75 [52.60, 58.98]	55.18 [51.51, 58.62]
3-mer logistic	57.79 [54.41, 61.08]	56.50 [53.01, 59.94]
6-mer logistic	59.95 [56.54, 63.17]	54.47 [50.87, 57.68]
Frozen DNABERT	56.44 [52.99, 59.63]	56.16 [52.43, 59.57]
Fine-tuned DNABERT	55.68 [52.34, 59.05]	53.31 [49.98, 56.74]

Performance across split settings

Fine-tuned DNABERT achieved 55.68% mean balanced accuracy on stratified splits and 53.31% on grouped splits (Table 1; Figure 1). Its fold standard deviations were 5.16 and 3.60 percentage points, respectively; these describe fold dispersion rather than confidence intervals. The grouped conditional interval, 49.98–56.74%, included the 50% balanced-accuracy reference.

The highest observed stratified score was 59.95% for 6-mer logistic regression. On grouped splits, 3-mer logistic regression had the highest observed score at 56.50%, followed by frozen DNABERT at 56.16%. The ordering is descriptive; the same outer predictions cannot serve as an independent dataset for selecting and then unbiasedly evaluating a winning method. The class-prior predictor obtained 50% balanced accuracy in both settings.

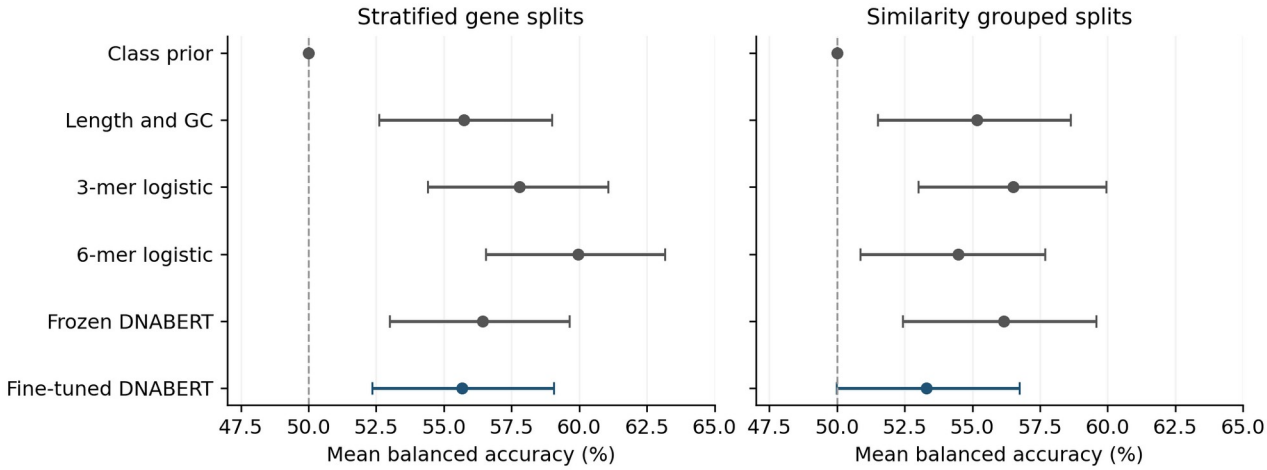


Figure 1. Mean outer-fold balanced accuracy and conditional 95% bootstrap intervals for six workflows. The dashed line marks 50% balanced accuracy. Blue points identify fine-tuned DNABERT. Intervals are conditional on fitted models and are not intervals for repeated training runs.

Paired differences and secondary metrics

Fine-tuning had lower point estimates than frozen DNABERT and each nonconstant classical baseline in both settings. Fine-tuning minus frozen performance was −0.75 percentage points on stratified splits and −2.85 points on grouped splits. Fine-tuning minus the best observed classical comparator was −4.27 points for stratified 6-mers and −3.19 points for grouped 3-mers. All primary balanced-accuracy intervals in Table 2 included zero. These results establish neither superiority nor equivalence. The upper bound for the stratified 6-mer comparison was very close to zero, so additional precision is retained in that table.

Table 2. Fine-tuned DNABERT minus comparator in percentage points of mean outer-fold balanced accuracy. Negative values favor the comparator.

Comparator	Stratified difference [95% interval]	Grouped difference [95% interval]
Frozen DNABERT	-0.75 [-4.522, +3.315]	-2.85 [-6.848, +1.658]
3-mer logistic	-2.11 [-5.831, +1.763]	-3.19 [-7.016, +0.529]
6-mer logistic	-4.27 [-8.412, +0.002]	-1.16 [-5.634, +3.488]
Length and GC	-0.07 [-3.620, +3.599]	-1.87 [-5.489, +1.868]

Fine-tuning pooled AUROC was 0.565 for stratified splits and 0.548 for grouped splits. The corresponding frozen values were 0.593 and 0.583. The 6-mer baseline achieved the highest stratified AUROC, 0.643; the 3-mer baseline achieved the highest grouped AUROC, 0.595. These secondary results did not suggest that the fine-tuned workflow’s balanced-accuracy result concealed a strong ranking advantage (Table 3). Pooled AUROC additionally mixes probabilities from different outer models, so its interpretation differs from the mean-fold primary endpoint.

Table 3. Secondary pooled outer out-of-fold metrics. These are point estimates, not independent replications.

Model	Stratified AUROC	Grouped AUROC	Stratified MCC	Grouped MCC
Length and GC	0.578	0.569	0.118	0.103
3-mer logistic	0.585	0.595	0.155	0.132
6-mer logistic	0.643	0.590	0.198	0.089
Frozen DNABERT	0.593	0.583	0.129	0.123
Fine-tuned DNABERT	0.565	0.548	0.114	0.065

Discussion

The central finding is a lack of demonstrated benefit from this specific DNABERT fine-tuning workflow over inexpensive CDS composition features. All intended fits completed and recorded encoder hashes changed during training. Encoder updates alone do not demonstrate useful adaptation. The held-out predictions provide the relevant evidence, and they did not establish an advantage on either evaluation setting.

The simple baselines help interpret the result. Length/GC performance above the constant reference suggests that basic sequence properties carry some information about the curated labels. The stronger k-mer scores show that short-pattern frequencies can be competitive without contextual encoder adaptation. They do not establish which biological mechanisms explain the association. Composition may reflect several correlated properties of the selected gene set, and no causal or mechanistic analysis was performed.

The lower fine-tuning point estimate under grouped evaluation is consistent with a more difficult generalization challenge, but the two designs use different held-out assignments and training compositions. Their score difference cannot be interpreted as a direct measurement of homology-induced inflation. Grouping only guarantees isolation of components in the declared detected-protein-similarity graph. Remote relationships, shared domains below the coverage criterion, and similarities outside coding proteins remain possible.

The training budget is a major constraint on interpretation. One learning rate and at most two inner epochs do not characterize the best achievable DNABERT performance. Selection of the second epoch in 21 of 30 inner fits indicates that the selected epoch often reached the budget boundary; it does not establish convergence or prove that more epochs would improve generalization. Longer training, alternative learning rates, and repeated seeds require a separately specified experiment. Once these outer scores have been inspected, further choices guided by them should be described as adaptive development rather than a fresh untouched evaluation.

Representation choices may also matter. Length-weighted mean pooling gives each gene one supervised loss and retains every CDS base, but it discards inter-window attention and global order. Complete CDS omits promoters, introns, and other regulatory context. These limitations offer plausible questions for follow-up work, not explanations demonstrated by this experiment. The planned matched-cohort CDS/promoter/genomic comparison has not been run and contributes no evidence to the present findings.

The study’s contribution is an auditable comparison rather than a new architecture or a clinically validated predictor. Fixed folds, source and checkpoint hashes, preserved training IDs, gene-level pooling, and paired prediction evidence make the result inspectable. That auditability is useful even when the more expensive

model does not lead the benchmark. It does not replace independent reproduction or establish that every upstream annotation and grouping choice is biologically complete.

Limitations

The cohort is small, historically curated, and restricted to exclusively annotated roles. Dual-role, unresolved, and unannotated genes were excluded, so performance cannot be generalized to all human genes or context-dependent role assignment. Unknown original export releases limit temporal interpretation. The union of sources may contain annotation and ascertainment differences; a consensus-only sensitivity analysis was not performed. The reference transcript policy is consistent but not tissue-specific, and only one transcript was modeled per gene.

Only one fine-tuning seed, one learning rate, one pooling method, and a short epoch budget were evaluated. Computational resources and tuning opportunities were not equalized across model families. Conditional bootstrap intervals omit retraining and split-construction variability and do not solve dependence created by overlapping training folds. No external independent cohort, prospective task, mechanistic validation, or deployment evaluation was included. Full learned checkpoints were absent from the submitted lightweight result archive, although checkpoint validation and fit records were retained. Numerical portability beyond the recorded MPS environment was not demonstrated.

Conclusions

In this 809-gene CDS benchmark, the specified short DNABERT fine-tuning protocol did not establish an advantage over frozen DNABERT or simple length/composition classifiers. Grouped evaluation yielded a fine-tuned balanced-accuracy interval that included 50%. These findings support a narrow conclusion about the tested workflow and emphasize the value of explicit baselines, gene-level fitting boundaries, and similarity-aware evaluation. Broader tuning, repeated-seed evaluation, matched sequence-input comparisons, and independent data are needed before making stronger claims about model-family performance or biological utility.

Declarations

Data and code availability

The project repository is https://github.com/Zelun-He/gene_classifier_project. Reference methods and frozen split records are identified by commit 132253d7557c87db2c755b6f85a7078005029490. The completed Mac run is identified by f725005e5126dcf152bbca9bb21c1b7644ef9cbf2014640f60a13ad69a3f1afa. Its lightweight archive contains predictions, fit and partition records, selection evidence, metrics, runtime provenance, and audit records. The new Mac launcher and completed run artifacts have not yet been deposited in a public research archive. A permanent public accession and an executable reproducibility release must be added before submission, subject to the source databases' redistribution terms. No public DOI is claimed here.

Ethics and consent

The analysis used reference sequences and curated gene annotations. No participants were recruited and no individual-level clinical records were analyzed. [The author must confirm the applicable institutional or journal ethics statement; no formal exemption determination is asserted.]

Funding competing interests and author contributions

[Funding sources, competing interests, the final author list and affiliations, and CRediT contributions require author confirmation. No statements of absent funding or absent conflicts are assumed. Contributions and acknowledgement wording concerning Dr. Hui Liu require agreement before submission.]

Use of generative artificial intelligence

Generative AI assisted with software development, analysis support, and manuscript drafting. The author is responsible for reviewing the implementation, results, citations, and final text. [Confirm the tools, versions, extent of use, and target journal disclosure requirements before submission.]

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Supplementary methods and results

Outer fold fine tuning details

Table S1. Fine-tuned outer-fold balanced accuracy, refit duration, and inner-selected decision threshold. Indices are zero-based.

Setting	Fold	Balanced accuracy %	Refit epochs	Threshold
Stratified	0	50.62	2	0.524430
Stratified	1	58.71	2	0.521269
Stratified	2	55.14	2	0.488490
Stratified	3	51.12	2	0.534890
Stratified	4	62.81	1	0.513798
Grouped	0	50.34	2	0.502527
Grouped	1	54.57	1	0.495309
Grouped	2	58.83	2	0.464852
Grouped	3	52.76	2	0.504425
Grouped	4	50.05	2	0.491030

Source and run identifiers

Recovered COSMIC export SHA-256: 5f5fbe6e68533e9f771680eb1473f6242913993f2429891c9bf48af363b7d9de. Recovered OncoKB export SHA-256: 802e957942bbfbe73127824e9b1841a4973bfa978741f81b99bd8ec4353f3994. Original release labels and retrieval dates remain unknown.

Frozen split SHA-256: 74ff6611274081d3aea28e1888d21fe5541145f99d90d0d4b6ec8f7bd43f7bdf. Mac launcher SHA-256: 54aebca1746a3e4afc697107b51d4fb54bb442d6144a92f11a268dd4595eb284. The run manifest binds 15 saved artifacts by SHA-256, including the outer prediction table, fit evidence, partition records, selections, summary, and tokenization audit.

Verification scope

The reported verification checks operated on the submitted result archive and the recorded reference methods. They confirmed internal consistency and the stated saved partition boundaries. They did not independently regenerate the biological annotations, reference extraction, or MMseqs2 alignments, and did not replay all 40 neural fits. The 4,389 tokenization records and per-gene prediction records remain the supporting evidence for complete input coverage and gene-level scoring.